# Week 13: Activity

## Mitchell Levy

## Data Visualization

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

```
--- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 ---
v forcats  1.0.1     v stringr  1.5.1
v ggplot2  3.5.1     v tibble   3.2.1
v lubrideate 1.9.4   v tidyr    1.3.1
v purrr    1.0.2
--- Conflicts ------------------------------------------ tidyverse_conflicts() ---
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to b
```

```
epl <- read_csv("epl_training.csv")
```

```
Rows: 6121 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (4): Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl  (18): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HC, AC, HF, AF, HY, AY, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
epl
```

| Date | HomeTeam | AwayTeam | FTHG | FTAG | FTR | HTHG | HTAG | HTR | Referee | HS | AS | HST | AST | HC | AC | HF | AF | HY | AY | HR | AR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 16/08/2015 | Chelsea | Burnley | 3 | 0 | H | 1 | 0 | H | M Oliver | 17 | 8 | 7 | 2 | 7 | 1 | 12 | 21 | 2 | 1 | 0 | 0 |
| 16/08/2015 | Charlton | Man City | 0 | 2 | A | 0 | 1 | A | J Moss | 8 | 17 | 2 | 5 | 6 | 6 | 23 | 16 | 2 | 0 | 0 | 0 |
| 16/08/2015 | Chelsea | West Ham | 4 | 2 | H | 1 | 0 | H | K Friend | 16 | 8 | 6 | 3 | 6 | 2 | 19 | 12 | 1 | 1 | 0 | 0 |
| 16/08/2015 | Coventry | Middlesbrough | 1 | 3 | A | 1 | 1 | D | L Mason | 13 | 14 | 4 | 5 | 7 | 4 | 15 | 15 | 3 | 0 | 0 | 0 |
| 16/08/2015 | Leicester | Tottenham | 2 | 0 | H | 0 | 0 | D | K Stroud | 6 | 13 | 2 | 3 | 5 | 8 | 12 | 19 | 2 | 0 | 0 | 0 |
| 16/08/2015 | Liverpool | Stoke | 1 | 0 | H | 1 | 0 | H | A Taylor | 19 | 10 | 7 | 2 | 10 | 5 | 12 | 12 | 3 | 1 | 0 | 0 |
| 16/08/2015 | Leicester | Aston Villa | 0 | 2 | A | 0 | 0 | D | N Swarbrick | 7 | 10 | 3 | 5 | 6 | 5 | 16 | 23 | 1 | 0 | 0 | 0 |
| 16/08/2015 | Sunderland | Arsenal | 0 | 3 | A | 0 | 1 | A | M Dean | 11 | 16 | 3 | 6 | 3 | 6 | 19 | 11 | 1 | 0 | 0 | 0 |
| 16/08/2015 | Swansea | Southampton | 1 | 0 | H | 1 | 0 | H | D Coote | 9 | 14 | 3 | 4 | 5 | 7 | 11 | 14 | 2 | 0 | 0 | 0 |
| 16/08/2015 | Crystal Palace | Man City | 0 | 3 | A | 0 | 2 | A | M Atkinson | 8 | 18 | 3 | 7 | 4 | 7 | 14 | 11 | 2 | 0 | 0 | 0 |

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H2. Total number of corners by the home team AC. Total number of corners by the away team

FTHG. The goals scored by the home team at full time FTAG. The goals scored by the away team at full time

HS. Total number of shots on goal by the home team AS. Total number of shots on goal by the away team

AR. Total number of red cards received by the away team

## 1. What is the impact of corner kicks on eventual goals (whole league or one team?) in a game or season (choose your timeframe)?

Angle: I am going to make a figure with 3 subplots, one for each year showing Liverpools corner kicks and total goals scored.

First, we need to get the data with a filter put (per hompage)

```
epl_1
epl_liverpool <- epl_1[epl_1$HomeTeam == "Liverpool" | epl_1[epl
epl_liverpool
```

| Date | HomeTeam | AwayTeam | FTHG | FTAG | FTR | HTHG | HTAG | HTR | Referee | HS | AS | HST | AST | HC | AC | HF | AF | HY | AY | HR | AR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 16/08/2015 | Liverpool | Bradford | 1 | 0 | H | 0 | 0 | D | S Attwell | 17 | 11 | 7 | 3 | 8 | 3 | 13 | 11 | 1 | 1 | 0 | 0 |
| 23/08/2015 | Arsenal | Liverpool | 0 | 2 | A | 0 | 1 | A | M Jones | 13 | 12 | 4 | 5 | 5 | 6 | 11 | 11 | 2 | 0 | 0 | 0 |
| 26/09/2015 | Southampton | Liverpool | 2 | 3 | A | 1 | 1 | D | L Mason | 12 | 17 | 5 | 7 | 4 | 7 | 11 | 11 | 2 | 0 | 0 | 0 |
| 03/10/2015 | Liverpool | Aston Villa | 1 | 0 | H | 1 | 0 | H | M Atkinson | 14 | 8 | 5 | 2 | 7 | 3 | 11 | 14 | 1 | 0 | 0 | 0 |
| 17/10/2015 | West Ham | Liverpool | 2 | 1 | H | 1 | 1 | D | M Dean | 14 | 11 | 6 | 4 | 5 | 4 | 12 | 11 | 2 | 0 | 0 | 0 |
| 24/10/2015 | Liverpool | Sunderland | 1 | 2 | H | 1 | 0 | H | C Pawson | 17 | 13 | 7 | 5 | 7 | 3 | 11 | 13 | 1 | 0 | 0 | 0 |
| 31/10/2015 | Man United | Liverpool | 1 | 2 | A | 1 | 1 | D | M Oliver | 11 | 13 | 4 | 6 | 3 | 5 | 12 | 11 | 2 | 0 | 0 | 0 |
| 07/11/2015 | Liverpool | Everton | 1 | 1 | D | 1 | 1 | D | J Moss | 13 | 11 | 5 | 4 | 6 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |
| 21/11/2015 | Liverpool | Leicester | 1 | 1 | D | 1 | 0 | H | M Atkinson | 13 | 11 | 5 | 3 | 7 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |
| 28/11/2015 | Derby | Liverpool | 0 | 4 | A | 0 | 1 | A | M Dean | 11 | 16 | 3 | 8 | 3 | 7 | 11 | 11 | 2 | 0 | 0 | 0 |
| 12/12/2015 | Liverpool | Tottenham | 1 | 1 | D | 1 | 1 | D | M Atkinson | 13 | 11 | 5 | 4 | 6 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |

1-10 of 380 rows | 1-18 of 22 columns  Previous 1 2 3 4 5 6 ... 38 Next

Now we need to make a new and tele column, one for liverpool' corner kicks, and one for total goals scored.

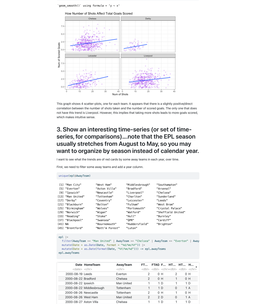
```
epl_liver <-
epl_liverpool$corners = epl_liverpool$HC
epl_liverpool$total_goals = epl_liverpool$FTHG + epl_liverpool$FTAG
epl_liverpool$corners
epl_liver
```

| Date | HomeTeam | AwayTeam | FTHG | FTAG | FTR | HTHG | HTAG | HTR | Referee | HS | AS | HST | AST | HC | AC | HF | AF | HY | AY | HR | AR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 16/08/2015 | Liverpool | Bradford | 1 | 0 | H | 0 | 0 | D | S Attwell | 17 | 11 | 7 | 3 | 8 | 3 | 13 | 11 | 1 | 1 | 0 | 0 |
| 23/08/2015 | Arsenal | Liverpool | 0 | 2 | A | 0 | 1 | A | M Jones | 13 | 12 | 4 | 5 | 5 | 6 | 11 | 11 | 2 | 0 | 0 | 0 |
| 26/09/2015 | Southampton | Liverpool | 2 | 3 | A | 1 | 1 | D | L Mason | 12 | 17 | 5 | 7 | 4 | 7 | 11 | 11 | 2 | 0 | 0 | 0 |
| 03/10/2015 | Liverpool | Aston Villa | 1 | 0 | H | 1 | 0 | H | M Atkinson | 14 | 8 | 5 | 2 | 7 | 3 | 11 | 14 | 1 | 0 | 0 | 0 |
| 17/10/2015 | West Ham | Liverpool | 2 | 1 | H | 1 | 1 | D | M Dean | 14 | 11 | 6 | 4 | 5 | 4 | 12 | 11 | 2 | 0 | 0 | 0 |
| 24/10/2015 | Liverpool | Sunderland | 1 | 2 | H | 1 | 0 | H | C Pawson | 17 | 13 | 7 | 5 | 7 | 3 | 11 | 13 | 1 | 0 | 0 | 0 |
| 31/10/2015 | Man United | Liverpool | 1 | 2 | A | 1 | 1 | D | M Oliver | 11 | 13 | 4 | 6 | 3 | 5 | 12 | 11 | 2 | 0 | 0 | 0 |
| 07/11/2015 | Liverpool | Everton | 1 | 1 | D | 1 | 1 | D | J Moss | 13 | 11 | 5 | 4 | 6 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |
| 21/11/2015 | Liverpool | Leicester | 1 | 1 | D | 1 | 0 | H | M Atkinson | 13 | 11 | 5 | 3 | 7 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |
| 28/11/2015 | Derby | Liverpool | 0 | 4 | A | 0 | 1 | A | M Dean | 11 | 16 | 3 | 8 | 3 | 7 | 11 | 11 | 2 | 0 | 0 | 0 |
| 12/12/2015 | Liverpool | Tottenham | 1 | 1 | D | 1 | 1 | D | M Atkinson | 13 | 11 | 5 | 4 | 6 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |

1-10 of 380 rows | 1-18 of 22 columns  Previous 1 2 3 4 5 6 ... 38 Next

Now we can graph

```
liver_data
epl_liverpool$corners = epl_liverpool$HC
epl_liverpool$total_goals = epl_liverpool$FTHG + epl_liverpool$FTAG
ggplot(epl_liverpool, aes(x = corners, y = total_goals))
+ geom_point() + facet_wrap(~ season) + labs(title = "How Liverpool's Corner Kicks Affect Total Goals Scored", x = "Number of Corner Kicks", y = "Total of Scored Goals")
liver_data
```




The figure shows the relationship between corner shots and number of goals scored by year for the team Liverpool. Surprisingly, there appears to be no relationship between these two variables. The points seem to be randomly distributed and magnitudes each year. Obviously 2016 has the most amount of data, but I think this is a decent graph.

## 2. What is the impact of shots on eventual goals?

I will make 3 graphs, one for each home team (Chelsea, shows the relationship between shots and turn of goals.

First, we need to clean the data.

```
epl_2
epl_chelsea$total_goals = epl_chelsea$FTHG + epl_chelsea$FTAG
epl_chelsea
```

| Date | HomeTeam | AwayTeam | FTHG | FTAG | FTR | HT | HTHG | HTAG | HTR | Referee | HS | AS | HST | AST | HC | AC | HF | AF | HY | AY | HR | AR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 16/08/2015 | Chelsea | West Ham | 4 | 2 | H | 1 | 0 | 0 | H | K Friend | 16 | 8 | 6 | 3 | 6 | 2 | 19 | 12 | 1 | 1 | 0 | 0 |
| 16/08/2015 | Chelsea | Burnley | 3 | 0 | H | 1 | 0 | 0 | H | M Oliver | 17 | 8 | 7 | 2 | 7 | 1 | 12 | 21 | 2 | 1 | 0 | 0 |
| 29/08/2015 | Chelsea | Southampton | 2 | 2 | D | 1 | 1 | 0 | D | M Atkinson | 14 | 11 | 5 | 5 | 5 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |
| 19/09/2015 | Chelsea | Bradford | 2 | 0 | H | 1 | 0 | 0 | H | S Attwell | 14 | 11 | 5 | 3 | 5 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |
| 26/09/2015 | Chelsea | Arsenal | 2 | 2 | D | 1 | 1 | 0 | D | M Dean | 14 | 11 | 5 | 5 | 5 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |
| 03/10/2015 | Chelsea | Middlesbrough | 3 | 0 | H | 1 | 0 | 0 | H | J Moss | 14 | 11 | 5 | 3 | 5 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |
| 24/10/2015 | Chelsea | Leicester | 3 | 1 | H | 1 | 1 | 0 | D | M Oliver | 14 | 11 | 5 | 4 | 5 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |
| 28/10/2015 | Chelsea | Aston Villa | 2 | 0 | H | 1 | 0 | 0 | H | M Atkinson | 14 | 11 | 5 | 3 | 5 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |
| 07/11/2015 | Chelsea | Sunderland | 3 | 1 | H | 1 | 1 | 0 | D | C Pawson | 14 | 11 | 5 | 4 | 5 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |
| 21/11/2015 | Chelsea | Liverpool | 2 | 1 | H | 1 | 1 | 0 | D | M Dean | 14 | 11 | 5 | 4 | 5 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |
| 28/11/2015 | Chelsea | Everton | 2 | 1 | H | 1 | 1 | 0 | D | J Moss | 14 | 11 | 5 | 4 | 5 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |
| 12/12/2015 | Chelsea | Man City | 2 | 1 | H | 1 | 1 | 0 | D | M Atkinson | 14 | 11 | 5 | 4 | 5 | 4 | 11 | 11 | 2 | 0 | 0 | 0 |

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Now let's graph it

```
epl_chelsea
ggplot(epl_chelsea, aes(x = HS, y = FTHG))
+ geom_point() + facet_wrap(~ season) + labs
(title = "Number of Shots Affect Total Goals Scored",
x = "Number of Shots", y = "Total of Scored Goals")
chelsea_data
```

```
geom_smooth() using formula = 'y ~ x'
```

The graph shows 3 scatter plots, one for each team. It appears that there is a slightly positive/linear correlation between the number of shots taken and the number of scored goals. The only one that does not have this trend is Liverpool. However, this implies that taking more shots leads to more goals scored, which seems intuitive.

## 3. Show an interesting time-series (or set of time-series, for comparisons)...note that the EPL season usually stretches from August to May, so you may want to organize by season instead of calendar year.

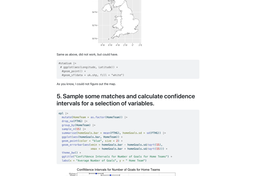
I want to see what the trends are of red cards by cross-daily teams in each year, over time.

First, we need to filter some away teams and add a year column.

```
library(dplyr)
```

```
epl_away_teams <-
group_by(AwayTeam) %>%
mutate(FTAR = if(FTR[n()] == 'A', 1L, 0L)) > Rdd 1L
mutate(Year = if(Date[n()] %in% c('2016', '2017', '2018')
Date[n()] %in% c('2016', '2017', '2018')
ungroup() %>% filter(!is.na(Year))
cat("Number of Fouls by Some Away Teams")
table(epl_away_teams$AwayTeam)
```

```
Warning: Removed 215 rows containing missing values or values outside the scale range
(`geom_bar()`).
```




This faceted figure shows 4 scatter plots, one for each team (Home). The scatter plot shows a time series of the number of fouls that each team committed over time. As you can see, all of the teams have a fairly similar number of fouls. Interestingly enough, all of the teams have committed more fouls in similar ranges of decreasing in the middle of the data, and then they come back up and then back down.

## 4. Unfortunately this data set doesn't lend itself well to a map, but there is no location data. Can you find the location of certain Premier League stadiums and make a map of UK to show some data (total wins, red cards?)

Note I could not get 4. I am leaving in all of my broken code for it, because it still could be useful to look at in the future.

```
library(dplyr)
library(maps)
```

```
Attaching package: 'maps'

The following object is masked from 'package:purrr':

    map
```

```
library(ggmap)
```

```
Loading to 2025-3-3.4, GDAL 3.5.3, PROJ 9.3.0, sf_use_s2() is TRUE
```

```
library(dplyr)
```

```
epl
```

| Date | HomeTeam | AwayTeam | FTHG | FTAG | FTR | HT | HTHG | HTAG | HTR | Referee | HS | AS | HST | AST | HC | AC | HF | AF | HY | AY | HR | AR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 16/08/2015 | Chelsea | Burnley | 3 | 0 | H | 1 | 0 | 0 | H | M Oliver | 17 | 8 | 7 | 2 | 7 | 1 | 12 | 21 | 2 | 1 | 0 | 0 |
| 16/08/2015 | Charlton | Man City | 0 | 2 | A | 0 | 1 | 0 | A | J Moss | 8 | 17 | 2 | 5 | 6 | 6 | 23 | 16 | 2 | 0 | 0 | 0 |
| 16/08/2015 | Chelsea | West Ham | 4 | 2 | H | 1 | 0 | 0 | H | K Friend | 16 | 8 | 6 | 3 | 6 | 2 | 19 | 12 | 1 | 1 | 0 | 0 |
| 16/08/2015 | Coventry | Middlesbrough | 1 | 3 | A | 1 | 1 | 0 | D | L Mason | 13 | 14 | 4 | 5 | 7 | 4 | 15 | 15 | 3 | 0 | 0 | 0 |
| 16/08/2015 | Leicester | Tottenham | 2 | 0 | H | 0 | 0 | 0 | D | K Stroud | 6 | 13 | 2 | 3 | 5 | 8 | 12 | 19 | 2 | 0 | 0 | 0 |
| 16/08/2015 | Liverpool | Stoke | 1 | 0 | H | 1 | 0 | 0 | H | A Taylor | 19 | 10 | 7 | 2 | 10 | 5 | 12 | 12 | 3 | 1 | 0 | 0 |
| 16/08/2015 | Leicester | Aston Villa | 0 | 2 | A | 0 | 0 | 0 | D | N Swarbrick | 7 | 10 | 3 | 5 | 6 | 5 | 16 | 23 | 1 | 0 | 0 | 0 |
| 16/08/2015 | Sunderland | Arsenal | 0 | 3 | A | 0 | 1 | 0 | A | M Dean | 11 | 16 | 3 | 6 | 3 | 6 | 19 | 11 | 1 | 0 | 0 | 0 |
| 16/08/2015 | Swansea | Southampton | 1 | 0 | H | 1 | 0 | 0 | H | D Coote | 9 | 14 | 3 | 4 | 5 | 7 | 11 | 14 | 2 | 0 | 0 | 0 |
| 16/08/2015 | Crystal Palace | Man City | 0 | 3 | A | 0 | 2 | 0 | A | M Atkinson | 8 | 18 | 3 | 7 | 4 | 7 | 14 | 11 | 2 | 0 | 0 | 0 |

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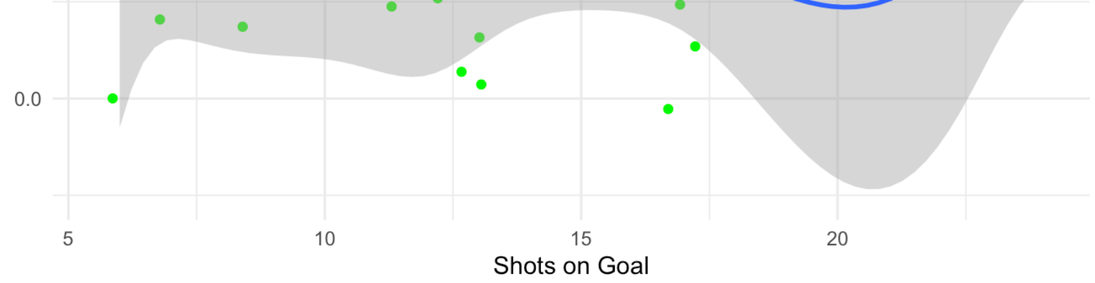
```
epl_liverpool$AwayTeam
```

```
[1] "Man City"      "Man City"      "West Ham"      "Tottenham"
[5] "Stoke"         "Aston Villa"   "Arsenal"       "Southampton"
[9] "Man City"      "Bournemouth"   "Bournemouth"   "Sunderland"
[13] "Tottenham"     "Bournemouth"   "Man City"      "Man City"
[17] "Tottenham"     "Tottenham"     "Man City"      "Tottenham"
[21] "Man City"      "Man City"      "Man City"      "Man City"
[25] "Man City"      "Man City"      "Man City"      "Man City"
[29] "Man City"      "Man City"      "Man City"      "Man City"
[33] "Man City"      "Man City"      "Man City"      "Man City"
```




Looking at this graph the blue line represents the model's prediction for what Chelsea's total goals scored will be depending on the shots on goal. The grey confidence band represents the uncertainty of where the model will be on the graph. If you made a 100 models of this graph using this sample, then 95 of the blue lines would be in this confidence band. As you can see, it seems that there is a sweet spot for the amount of shots on goal that a team should have, which is around 15-16 shots. This is the peak of the graph, and any more or less than that and the amount of goals scored goes down in either direction.